

The empirical recurrence for OEIS A236784: recovering a conjecture that is not written down

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Abstract

OEIS A236784 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 207$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236784 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the upper median plus the lower median minus the minimum of every 2×2 subblock equal”. It has offset 1 and begins

770, 4452, 23675, 145124, 867928, 5547565, 35115362, 230220290, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:32:39, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 207$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 207$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 414$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	28	c_{22}	15143385673379179	c_{43}	153797473431616303838547	c_{64}	13685459
c_2	−138	c_{23}	233451188720055518	c_{44}	2755242749977941205912	c_{65}	−821383
c_3	−3131	c_{24}	−174825791573522016	c_{45}	−254779186886689729303186	c_{66}	−4260328
c_4	35138	c_{25}	−1767652157336689202	c_{46}	−34481742380708237124529	c_{67}	−209934
c_5	98877	c_{26}	1467431964468868485	c_{47}	359745868383003112593830	c_{68}	10489376
c_6	−2800396	c_{27}	11407789258412574883	c_{48}	88890095748878826722233	c_{69}	1541287
c_7	3228326	c_{28}	−9623770890253871944	c_{49}	−431313002884093671200740	c_{70}	−198531
c_8	125292435	c_{29}	−63127434706444524945	c_{50}	−153049379398767957994553	c_{71}	−482044
c_9	−409487545	c_{30}	50770128670992361147	c_{51}	437009949663734140734854	c_{72}	2748730
c_{10}	−3658343046	c_{31}	300650993083797005380	c_{52}	201687752489851679515320	c_{73}	972787
c_{11}	18570209996	c_{32}	−218117763864362502635	c_{53}	−371871546416467790083348	c_{74}	−25050
c_{12}	73890433500	c_{33}	−1234668756586394898319	c_{54}	−212599809044824918936599	c_{75}	−13309
c_{13}	−537953433565	c_{34}	764888665201445819942	c_{55}	263490591654515798611976	c_{76}	10321
c_{14}	−1036914148304	c_{35}	4373980839183196365696	c_{56}	182559643848927772206706	c_{77}	11897
c_{15}	11353606846264	c_{36}	−2173683372325258383932	c_{57}	−153490862386214424053150	c_{78}	53630
c_{16}	9262834011851	c_{37}	−13359164252517494081969	c_{58}	−128660670405546124340154	c_{79}	−6098
c_{17}	−185372213922877	c_{38}	4896179587664924611430	c_{59}	72006378868591688783024	c_{80}	−894
c_{18}	−25208328378633	c_{39}	35129294952621730751682	c_{60}	74532873527896704989144	c_{81}	1107
c_{19}	2425698908549131	c_{40}	−8242478561670752859501	c_{61}	−26172691324997580887576	c_{82}	343
c_{20}	−730637245299426	c_{41}	−79382846980586487706211	c_{62}	−35393400947227810195624	c_{83}	219
c_{21}	−26048490590020008	c_{42}	8389524523571666152049	c_{63}	6719942631127418614072		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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